Princeton University

ORF 363/COS 323
Final Exam, Fall 2022

December 18, 2022

Instructor:
I. Akrotirianakis

AIs:
A. Chaudhry, B. Strekha, Z. Li

1. Please write your name on the first page of your solutions. Next to it, please write out and sign the following pledge: "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this examination."
2. The exam is not to be discussed with *anyone* except possibly the instructors and the AIs. You can only ask *clarification questions*, and only as *public* (and preferably non-anonymous) questions on Ed Discussion. No emails.
3. You are allowed to consult the lecture notes, your own notes, the reference books of the course as indicated on the syllabus, the problem sets and their solutions (yours and ours), the midterm and its solutions (yours and ours), the practice midterm and final exams and their solutions, all Ed Discussion posts, but *nothing else*. You can only use the Internet in case you run into problems related to software.
4. You may refer to facts proven in the notes or problem sets without reproving them.
5. For all computational problems, include your code. The output that you present should come from your code.
6. Unless you have been granted an extension because of overlapping finals, the exam is to be turned in on Tuesday (December 20, 2022) at 10 AM EST on Gradescope as a single PDF file. You are free to write your solutions on paper or on a tablet, or to type them up. Only the latest version submitted before your deadline will be graded.
7. Each question has 20 points. You need to justify your answers to receive full credit.

Problem 1: Convexity

(a) Show that $x = (0, 0, 0)$ is a strict global minimum of the function:

$$f(x, y, z) = x^2 + y^2 + z^2 - xy + yz - xz. \tag{1}$$

(b) Consider the function

$$f(x, y) = e^{x-y} + e^{y-x}. \tag{2}$$

Show that all the points $(x, y) \in \mathbb{R}^2$ with $x = y$ are global minimizers of f (or equivalently, all the points on the line $x - y = 0$ are global minimizers).

(c) Let f be a function from $\mathbb{R}^n$ to $\mathbb{R}$. Recall an affine function is one that can be written as $f(x) = a^T x + b$ for some vector a and some scalar b. For the following statements, provide a proof or a counterexample.

(i) If f is both convex and concave, then f is affine.
Hint: You may assume f is twice differentiable.

(ii) If $g \circ f$ is convex for all convex functions g, then f is affine.

Solution:

(a) The gradient and Hessian are

$$\nabla f = \begin{bmatrix} 2x - y - z \\ 2y - x + z \\ 2z + y - z \end{bmatrix} \tag{3}$$

$$\nabla^2 f = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \tag{4}$$

By Sylvester's criterion, the Hessian is pd so the function is strictly convex. The gradient at $(0, 0, 0)$ is the zero vector. By strict convexity, $(0, 0, 0)$ is a strict global minimum of $f(x, y, z)$.

(b) The gradient and Hessian are

$$\nabla f = \begin{bmatrix} e^{x-y} - e^{y-x} \\ -e^{x-y} + e^{y-x} \end{bmatrix} \tag{5}$$

$$\nabla^2 f = \begin{bmatrix} e^{x-y} + e^{y-x} & -e^{x-y} - e^{y-x} \\ -e^{x-y} - e^{y-x} & e^{x-y} + e^{y-x} \end{bmatrix}. \tag{6}$$

The determinant of the Hessian is 0, and the diagonal elements are non-negative. By Sylvester's criterion, the Hessian is psd. For points with $x = y$, the gradient is the zero vector. Thus, the points on the line $x - y = 0$ are global minimizers.

(c) (i) True. We present two proofs, one using calculus, and one without it.

First proof:

By convexity (resp. concavity) we have $\nabla^2 f \succeq 0$ (resp. $\nabla^2 f \preceq 0$). Thus $\nabla^2 f \equiv 0$. It follows that the first-order Taylor approximation is exact. Thus $f(x) = f(0) + \nabla f(0)^T x$. Let $b = f(0)$ and $a = \nabla f(0)$.

Second proof:

Define $b = f(0)$. Let e_i be the vector with 1 as its ith entry and with the rest of its entries being 0. For $i = 1, \ldots, n$ define $a_i = f(e_i) - b$. We now claim $f(x) = a^T x - b$ for all $x \in \mathbb{R}^n$. We proceed by induction on the number of nonzero entries of x. The base case is when $x = 0$ and in this case $f(0) = a^T 0 + f(0) = a^T x + b$. If only one entry of x is nonzero then we can write $x = te_i$ for some nonzero $t \in \mathbb{R}$ and some i. If $t = 1$, then $f(x) = f(e_i) = f(e_i) - f(0) + f(0) = a_i + b = a^T x + b$. If $t > 1$, let $\lambda = \frac{1}{t}$ and observe by convexity and concavity:

$$\begin{aligned} f(\lambda x + (1-\lambda)0) &= \lambda f(x) + (1-\lambda) f(0) \\ f(e_i) &= \frac{1}{t} f(x) + (1 - \frac{1}{t}) b \\ t f(e_i) &= f(x) + (t-1) b \\ f(x) &= t(f(e_i) - b) + b \\ &= t a_i + b \\ &= a^T x + b. \end{aligned}$$

If $t \in (0, 1)$, let $\lambda = t$ and observe:

$$\begin{aligned} f(\lambda e_i + (1-\lambda)0) &= \lambda f(e_i) + (1-\lambda) f(0) \\ f(x) &= t f(e_i) + (1-t) b \\ &= t(f(e_i) - b) + b \\ &= t a_i + b \\ &= a^T x + b. \end{aligned}$$

If $t < 0$, let $\lambda = \frac{1}{1-t}$ and observe by convexity and concavity:

$$\begin{aligned}
f(\lambda x + (1-\lambda)e_i) &= \lambda f(x) + (1-\lambda) f(e_i) \\
f(0) &= \frac{1}{1-t} f(x) + \frac{-t}{1-t} f(e_i) \\
f(x) &= (1-t) f(0) + t f(e_i) \\
&= f(0) + t(f(e_i) - f(0)) \\
&= t a_i + b \\
&= a^T x + b.
\end{aligned}$$

Now we assume we have shown equality for all x with $k \geq 1$ nonzero entries. Take a vector x with $k+1$ nonzero entries. Let i be the index of one of the nonzero entries. Let $\tilde{x}$ be equal to x except with its ith entry replaced by 0. Note $x = \frac{1}{2}(2\tilde{x}) + \frac{1}{2}(2x_i e_i)$ and both $2\tilde{x}$ and $2x_i e_i$ have at most k nonzero entries, therefore by induction we have $f(2\tilde{x}) = a^T 2\tilde{x} + b$ and $f(2x_i e_i) = a^T 2x_i e_i + b$. Now by convexity and concavity:

$$\begin{aligned}
f(x) &= f(\frac{1}{2}(2\tilde{x}) + \frac{1}{2}(2x_i e_i)) \\
&= \frac{1}{2} f(2\tilde{x}) + \frac{1}{2} f(2x_i e_i) \\
&= \frac{1}{2}(a^T 2\tilde{x} + b) + \frac{1}{2}(a^T 2x_i e_i + b) \\
&= a^T(\frac{1}{2} 2\tilde{x} + \frac{1}{2} 2x_i e_i) + b \\
&= a^T x + b.
\end{aligned}$$

This completes the proof by induction.

(ii) True. Note that $g(x) = x$ and $g(x) = -x$ are convex functions, so $f(x)$ and $-f(x)$ must be convex which means that $f(x)$ must be convex and concave. By part (i), $f(x)$ is affine.

Problem 2: Nonlinear least squares

On a recent camping trip to Africa, our friend Alice was fascinated with the beauty of two flowers (let's call them A and B) that were growing close to her tent. She decided to collect measurements of their growth and use the knowledge she acquired during her college years to develop a mathematical model that fits the data well and then use it to predict the growth of the flowers. She collected measurement during certain days only, because she was busy exploring several villages and towns near the camp she was staying. The measurements for flowers A and B are as follows:

$$A = \{(1, 3.2939), (2, 4.2699), (4, 7.1749), (5, 9.3008), (8, 20.259), (10, 36.12)\},$$
$$B = \{(1, 3.023), (2, 4.36), (4, 6.258), (5, 8.21), (8, 17.85), (10, 21.25)\}.$$

The elements in the above datasets have the following structure (t, y), where t is the day Alice measured the flowers and y is the actual measurement.

Alice's friend John told her that she can try to fit an exponential growth model of the form

$$T(t) = c_1 e^{c_2 t}. \tag{7}$$

Alice recalled that she can use nonlinear least squares to determine the unknown parameters c_1 and c_2 in the model.

(a) Write the nonlinear least squares problem that minimizes the sum of the squares of the residuals between the data and the prediction from the growth model.

(b) Apply 15 iterations of the Gauss-Newton method to determine two separate models, one for flower A and another for flower B. For both models use the same starting point $(c_1^0, c_2^0) = (2.75, 0.275)$. Plot Alice's measurements (ground truth) and the best fit that you found on the same figure (one plot for plant A and another for plant B).

(c) Discuss the relation between the objective function value of nonlinear least squares and how well the resulting model fits the data.

Solution:

(a) Let y_t be the growth observed on day t. Then the nonlinear least squares problem we wish to solve is:

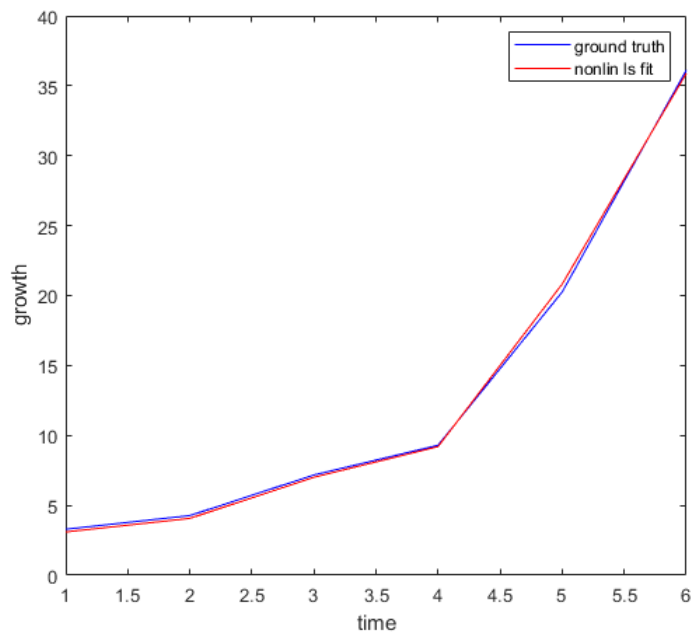
$$\min_{c_1, c_2 \in \mathbb{R}} \sum_t (c_1 e^{c_2 t} - y_t)^2.$$

(b) Here is code:

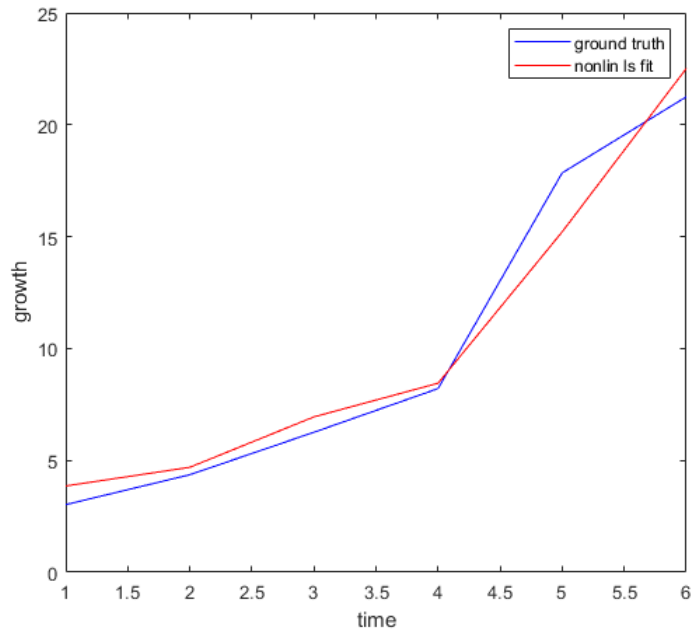
```
% number of iterations
iter=30;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%% DATA for Plant A %%%%%
xk=[2.75;0.275];
t=[1;2;4;5;8;10];
y=[3.2939;4.2699;7.1749;9.3008;20.259;36.12];

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%% DATA for Plant B %%%%%
% xk = [2.750;0.275];
% t = [1; 2; 4; 5; 8; 10];
% y = [3.023; 4.36; 6.258; 8.21; 17.85; 21.25];

m=length(y);

% the model we want to fit is: yi~=x1*exp(x2*ti)
gkfn = @(x1k,x2k) y - x1k*exp(x2k*t);

% gradient of the model wrt x1 and x2 (the unknown params)
jkfn = @(x1k,x2k) [-exp(x2k*t),-x1k*t.*exp(x2k*t)];

% Gauss-Newton method
for i=1:iter
    gk = gkfn(xk(1),xk(2));
    jk = jkfn(xk(1),xk(2));
    xk = xk - (jk'*jk)\(jk'*gk);
end

close all
q = [];
for i=1:m
    q(i)=xk(1)*exp(xk(2)*t(i));
```

```
end

time = 1:m;
plot(time, y, 'b')
hold on
plot(1:m, q, 'r')
legend('ground truth', 'nonlin ls fit')
xlabel('time')
ylabel('growth')
```

Here is the plot for flower A:



Here is the plot for flower B:



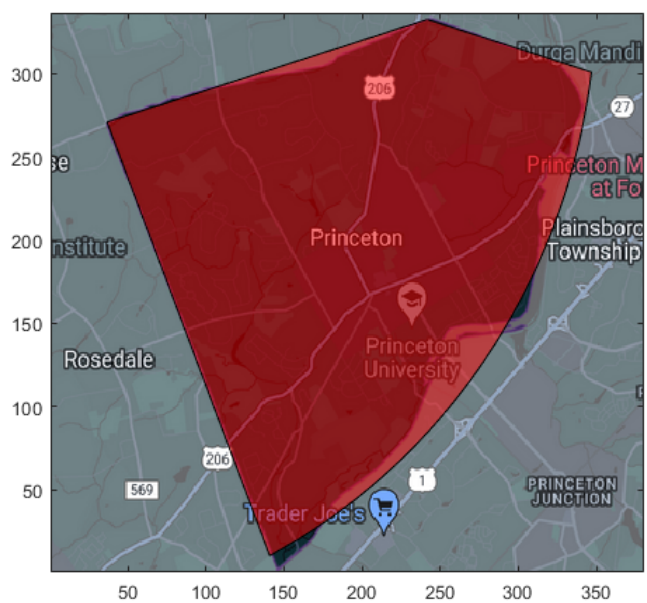
(c) The smaller the objective value, the better the fit. In this case, the objective value for flower A is very small indicating a good fit. On the other hand, the objective value for flower B is much larger, indicating a worse fit.

Problem 3: Radio Free Princeton

The town of Princeton is constructing an antenna to improve cell coverage. You have been hired to decide the antenna's location and range. The signal will reach all points whose euclidean distance to the antenna's location is less than the antenna's range. Due to possible interference with other towns' cell networks, you must ensure the signal only reaches points within the town's territory. Subject to this constraint, you want the signal to reach as large an area as possible. The town's territory can be described by the following convex set:

$$P = \left\{ x \in \mathbb{R}^2 \quad \text{s.t.} \quad \begin{array}{c} x_1 + 0.4x_2 \geq 145 \\ 0.3x_1 - x_2 >= -260 \\ 0.3x_1 + x_2 <= 405 \\ ||x - (-30, 350)|| <= 380 \end{array} \right\}.$$

The set is an intersection of three half-spaces (the first three constraints) and a circle (the last constraint.) As you can see below, this is actually just an approximation, but for the purposes of this problem, you can assume this is an exact description.



(a) First you have to do some theoretical work. Consider the circle $\mathcal{B}$ with center x_b and radius r_b defined as

$$\mathcal{B} = \{x \in \mathbb{R}^2 \mid ||x - x_b|| \leq r_b\}.$$

(i) Consider the half-space $\mathcal{H} = \{x \in \mathbb{R}^2 | a^T x \leq b\}$. Show that the constraint $\mathcal{B} \subseteq \mathcal{H}$ can be written (or expressed) as follows:

$$a^T x_b + r_b ||a|| \leq b.$$

Hint: $\mathcal{B} \subseteq \mathcal{H}$ is equivalent to $\max_{x \in \mathcal{B}} a^T x \leq b$

(ii) Consider another circle $\mathcal{C} = \{x \in \mathbb{R}^2 \mid ||x - x_c|| \leq r_c\}$. Show that the constraint $\mathcal{B} \subseteq \mathcal{C}$ can be written (or expressed) as follows:

$$||x_b - x_c|| + r_b \leq r_c.$$

Hint: $\mathcal{B} \subseteq \mathcal{C}$ is equivalent to $\max_{x \in \mathcal{B}} ||x - x_c|| \leq r_c$.

(b) Show that the optimal location and range can be found with a convex problem.

(c) Solve the convex problem using CVX and report the optimal location and range.

Solution:

(a) (i) Observe that $\mathcal{B} \subseteq \mathcal{H}$ is equivalent to $\max_{x \in \mathcal{B}} a^T x \leq b$. We just need to show $\max_{x \in \mathcal{B}} a^T x = a^T x_b + r_b ||a||$. First we show $\leq$. Observe for all $x \in \mathcal{B}$, by the Cauchy-Schwarz inequality

$$a^T x = a^T (x - x_b + x_b) = a^T x_b + a^T (x - x_b) \leq a^T x_b + ||a|| ||x - x_b|| \leq a^T x_b + r_b ||a||.$$

Now we show $\geq$. Take $x = x_b + r_b \frac{a}{||a||}$ which is clearly in $\mathcal{B}$. Observe

$$a^T x = a^T (x_b + r_b \frac{a}{||a||}) = a^T x_b + r_b \frac{a^T a}{||a||} = a^T x_b + r_b ||a||.$$

Thus we have $\max_{x \in \mathcal{B}} a^T x = a^T x_b + r_b ||a||$ as desired.

(ii) Observe that $\mathcal{B} \subseteq \mathcal{C}$ is equivalent to $\max_{x \in \mathcal{B}} ||x - x_c|| \leq r_c$. We just need to show $\max_{x \in \mathcal{B}} ||x - x_c|| = ||x_b - x_c|| + r_b$. First we show $\leq$. Observe for all $x \in \mathcal{B}$, by the triangle inequality

$$||x - x_c|| \leq ||x_b - x_c|| + ||x - x_b|| \leq ||x_b - x_c|| + r_b.$$

Now we show $\geq$. If $x_b \neq x_c$ we take $x = x_b + r_b \frac{x_b - x_c}{||x_b - x_c||}$ which is clearly in $\mathcal{B}$. Observe

$$||x - x_c|| = ||x_b + r_b \frac{x_b - x_c}{||x_b - x_c||} - x_c||$$

$$
\begin{aligned}
&= ||(x_b - x_c)(1 + \frac{r_b}{||x_b - x_c||})|| \\
&= ||x_b - x_c||(1 + \frac{r_b}{||x_b - x_c||}) \\
&= ||x_b - x_c|| + r_b.
\end{aligned}
$$

If $x_b = x_c$, let u be any unit vector and take $x = x_b + r_b u$. Then we have

$$||x - x_c|| = ||x - x_b|| = ||x_b + r_b u - x_b|| = ||r_b u|| = r_b||u|| = r_b = ||x_b - x_c|| + r_b.$$

Thus in either case we have $\max_{x \in \mathcal{B}} ||x - x_c|| \geq ||x_b - x_c|| + r_b$.

(b) Let x_a be the location of the antenna and let r_a be its range. The area the antenna's signal will reach is a circle which will have area πr_a^2. Therefore, to maximize the area, it is equivalent to maximize the radius, r_a. Let $B(x, r)$ be the circle centered at x with radius r. Then the problem we wish to solve is:

$$
\begin{aligned}
\max_{x_a \in \mathbb{R}^2, r_a \in \mathbb{R}} \quad & r_a \\
\text{s.t.} \quad & B(x_a, r_a) \subseteq P.
\end{aligned}
$$

For conciseness, let us rewrite P.

$$
P = \left\{ x \in \mathbb{R}^2 \quad \text{s.t.} \quad \begin{array}{c} a_1^T x \leq b_1 \\ a_2^T x \leq b_2 \\ a_3^T x \leq b_3 \\ ||x - x_4|| <= r_4 \end{array} \right\}
$$

For appropriately defined a_i, b_i, x_4 and r_4. For the circle $B(x_a, r_a)$ to be contained in P it is equivalent for the circle to be contained in each of the three halfspaces and the circle which respectively define P. So we can rewrite the problem:

$$
\begin{aligned}
\max_{x_a \in \mathbb{R}^2, r_a \in \mathbb{R}} \quad & r_a \\
\text{s.t.} \quad & B(x_a, r_a) \subseteq \{x \mid a_i^T x \leq b_i\} \quad i = 1, 2, 3 \\
& B(x_a, r_a) \subseteq \{x \mid ||x - x_4|| <= r_4\}.
\end{aligned}
$$

Using part (a) we can reformulate:

$$
\begin{aligned}
\max_{x_a \in \mathbb{R}^2, r_a \in \mathbb{R}} \quad & r_a \\
\text{s.t.} \quad & a_i^T x_a + r_a ||a_i|| \leq b_i \quad i = 1, 2, 3 \\
& ||x_a - x_4|| + r_a \leq r_4.
\end{aligned}
$$

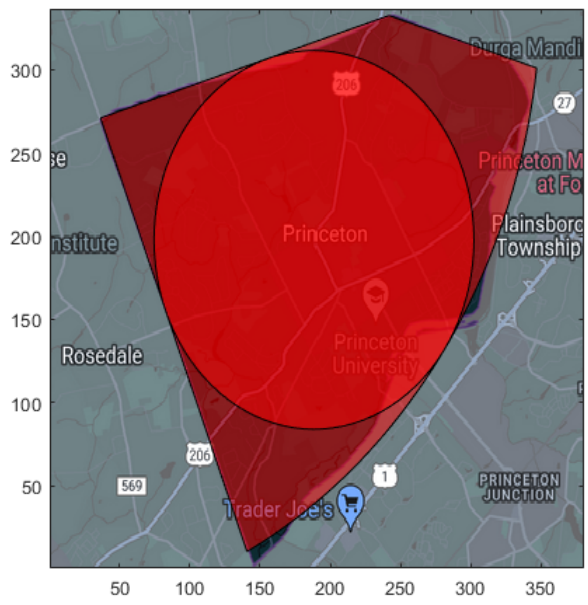
This is evidently a convex problem which we can put into CVX.

(c) We should place the antenna at $(188.4188, 197.7580)$ with range 113.7588.

Here is code that solves the problem with CVX:

```
a_1 = [-1;-.4];
b_1 = -145;
a_2 = [-.3;1];
b_2 = 260;
a_3 = [.3;1];
b_3 = 405;
x_4 = [-30;350];
r_4 = 380;

cvx_begin
variable x_a(2,1)
variable r_a
maximize r_a
subject to
a_1'*x_a + r_a*norm(a_1) <= b_1;
a_2'*x_a + r_a*norm(a_2) <= b_2;
a_3'*x_a + r_a*norm(a_3) <= b_3;
norm(x_a-x_4)+r_a <= r_4;
cvx_end
```

Here is a plot of the area that the signal reaches.


Durga Mandi
Princeton
Plainsboro
Township
Rosedale
Princeton
University
Trader Joe's
PRINCETON
JUNCTION
205
27
206
569
1
300
250
200
150
100
50
50
100
150
200
250
300
350

Problem 4: Miracle on 363rd Street

This holiday season, you are feeling generous and want to give away presents to the children in your neighborhood. There are n children so you have bought n presents. You will give one present to each child. All that is left is to decide which present to give to which child. Let E_{ij} be the enjoyment child i will receive from present j. You wish to maximize the total enjoyment of all the children.

(a) Formulate the problem of maximizing total enjoyment as an integer program.
Hint: Have a binary variable for each pair (i, j).

(b) Take the LP relaxation of your integer program.

(c) Using CVX, solve your derived LP with the matrix E provided in the file `Miracle.mat`.

(d) Report an optimal present distribution.

Solution:

(a)

$$\begin{aligned} \max_{X \in \mathbb{R}^{n \times n}} \quad & \sum_{ij} E_{ij} X_{ij} \\ \text{s.t.} \quad & \sum_{j} X_{ij} = 1 \quad i = 1, \ldots, n \\ & \sum_{i} X_{ij} = 1 \quad j = 1, \ldots, n \\ & X_{ij} \in \{0, 1\}. \end{aligned}$$

The variable X_{ij} is one if we give present j to child i and zero otherwise. The objective sums total enjoyment. The first constraint ensures each child gets exactly one present. The second constraint ensures each present is given to exactly one child.

(b)

$$\begin{aligned} \max_{X \in \mathbb{R}^{n \times n}} \quad & \sum_{ij} E_{ij} X_{ij} \\ \text{s.t.} \quad & \sum_{j} X_{ij} = 1 \quad i = 1, \ldots, n \\ & \sum_{i} X_{ij} = 1 \quad j = 1, \ldots, n \\ & 0 \leq X_{ij} \leq 1. \end{aligned}$$

(c) Here is code that solves the problem with CVX:

```
load("Miracle.mat")
n = length(E);
cvx_begin
variable X(n,n)
maximize (trace(E'*X))
subject to
0 <= X <= 1;
sum(X) == 1;
sum(X') == 1;
cvx_end
[r,c] = find(round(X));
for i=1:20
    fprintf('Child %d receives present %d. \n',r(i),c(i));
end
```

(d) The optimal solution to the LP is integral, so it is also optimal for the integer program. We get an optimal distribution by looking at the nonzero elements of the matrix X. Child 4 receives present 1. Child 16 receives present 2. Child 6 receives present 3. Child 10 receives present 4. Child 19 receives present 5. Child 14 receives present 6. Child 20 receives present 7. Child 15 receives present 8. Child 1 receives present 9. Child 12 receives present 10. Child 3 receives present 11. Child 11 receives present 12. Child 5 receives present 13. Child 17 receives present 14. Child 2 receives present 15. Child 18 receives present 16. Child 9 receives present 17. Child 7 receives present 18. Child 8 receives present 19. Child 13 receives present 20.

Problem 5: SDP

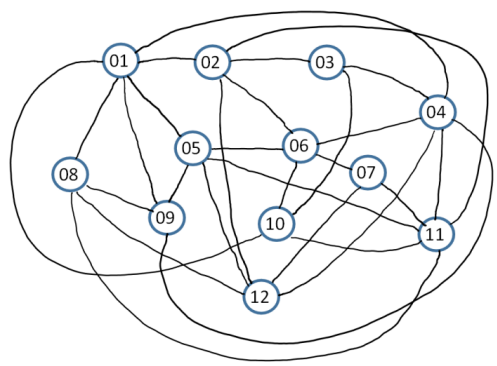
(a) **Conversion to SDP:** Consider the following optimization problem

$$\begin{aligned} \max_{x_1, x_2 \in \mathbb{R}} \quad & 3x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 10, \\ & x_1 x_2 \geq 4, \\ & x_1 + x_2 \geq 0. \end{aligned} \tag{8}$$

Write problem (8 as an SDP. Hint: consider the matrix

$$X = \begin{bmatrix} x_1 & 2 \\ 2 & x_2 \end{bmatrix}.$$

(b) **Let's finish where we started:** In the very first lecture, we tried to avoid "overlapping finals" by finding the largest stable set (aka independent set) in the following graph:



We found one of size 6, but at the time we couldn't show that 6 was the best possible without trying all possibilities. Solve a convex optimization problem (e.g., the LP relaxation) which proves that 6 is indeed optimal. Show your CVX code and its output. Argue precisely why the solution to your optimization problem certifies that 6 is the best possible. For your convenience, the adjacency matrix of the graph is provided below. You can copy and paste it into MATLAB.

$$G = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solution:

(a) Note that the objective and constraints

$$\begin{aligned} \max_{x_1, x_2 \in \mathbb{R}} \quad & 3x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 10, \\ & x_1 x_2 \geq 4, \\ & x_1 + x_2 \geq 0. \end{aligned} \tag{9}$$

can be rewritten as

$$\begin{aligned} \max_{x_1, x_2 \in \mathbb{R}} \quad & \operatorname{Tr}\left(\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 & 2 \\ 2 & x_2 \end{bmatrix} \right) \\ \text{s.t.} \quad & \operatorname{Tr}\left(\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 & 2 \\ 2 & x_2 \end{bmatrix} \right) \leq 10, \\ & \begin{bmatrix} x_1 & 2 \\ 2 & x_2 \end{bmatrix} \succeq 0, \\ & \operatorname{Tr}\left(\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 & 2 \\ 2 & x_2 \end{bmatrix} \right) \leq 0. \end{aligned} \tag{10}$$

(b) An LP bound of the problem returns and optimal value of 6 (see the code below). Since the upper bound to the maximum stable set is 6, and we have explicitly found a stable set of size 6, this means the bound is tight (6 is the best possible).

```
G =[
0 1 0 1 1 0 0 1 1 1 0 0;
1 0 1 0 0 1 0 0 0 0 1 1;
0 1 0 1 0 0 0 0 0 1 0 0;
1 0 1 0 0 1 0 0 1 0 1 1;
1 0 0 0 0 1 0 0 1 0 1 1;
0 1 0 1 1 0 1 0 0 1 0 0;
0 0 0 0 0 1 0 0 0 0 1 1;
1 0 0 0 0 0 0 0 1 0 1 1;
1 0 0 1 1 0 0 1 0 0 0 0;
1 0 1 0 0 1 0 0 0 0 1 0;
0 1 0 1 1 0 1 1 0 1 0 0;
0 1 0 1 1 0 1 1 0 0 0 0;
];

[~,N] = size(G);
cvx_begin
variable x(N)
maximize sum(x)
subject to
for i=1:N-1
    for j=i+1:N
        if G(i, j) == 1
            x(i) + x(j) <= 1;
        end
    end
end
x >= 0;
x <= 1;
cvx_end
```